

Lecture 2:

AI504 - Knowledge Representation

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The logic of “all p are q ”

- Fix a set (e.g. of nouns), e.g. {groundhogs, warthogs, hedgehogs}
- once we fix a set of nouns, we can form sentences of the form “all ___ are ___”
- These sentences are neither true nor false until we give them meaning (we don't necessarily know what a groundhog or a hedgehog is), until we interpret them with **model**.

Model

- A set of all underlying elements (*nouns*) in the univers/domain
- Then we can map each element to a ground in the set.

$\mathcal{M}$ usually refers to the model, while M refers to its domain

Lowercase greek letters like ϕ, ψ usually refers to sentences

Γ is a set of sentences

Then $\mathcal{M} \models \phi$, means that the model $\mathcal{M}$ satisfies the sentence ϕ .

Note that $\models$ is polymorphic where

$$\forall \psi \in \Gamma \mid \mathcal{M} \models \psi, \text{ if } \Gamma \models \phi \Rightarrow \mathcal{M} \models \phi$$

if any sentence in Γ satisfies the model $\mathcal{M}$ then if $\Gamma \models \phi$, then $\mathcal{M} \models \phi$

Definition of $\vdash$

For $\Gamma \vdash \phi$, means that there is a proof of ϕ from Γ

Example (*noun example*)

Let P be the underlying set of nouns. A P consists of of:

1. A set of X (the domain or universe)
2. A function $P \rightarrow \mathcal{P}(X)$ where $\mathcal{P}$ is the powerset of X

Now let

- all g are w
- all w are h
- all g are h

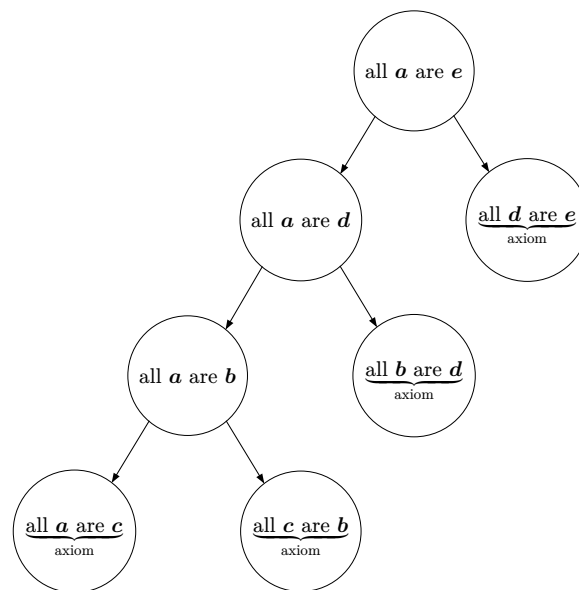
Note that these have the transitive property that $g \in w$ and $w \in h \Rightarrow g \in h$

Because of this the following statement is not possible

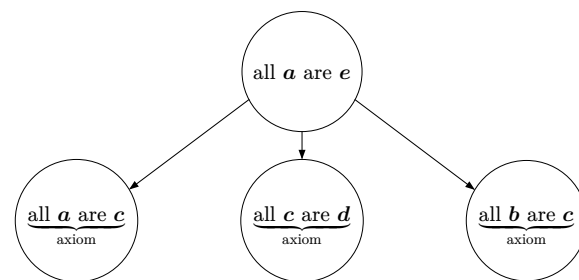
$$g \models w, \quad w \models h, \quad g \not\models h$$

let $\Gamma = \{(\text{all } a \text{ are } b), (\text{all } b \text{ are } d), (\text{all } b \text{ are } c), (\text{all } c \text{ are } b), (\text{all } d \text{ are } c), (\text{all } d \text{ are } e)\}$

Show that there are a “chain of reasoning” that uses three assumptions and deduces “all a are e ”



Induction on proof trees.



Is there a proof of “all a are e ” from only these 3 axioms?

No, since none of the axioms contain e

If a element appears in the conclusion of a proof, it must appear in one of the leaves.

We use Induction on proof trees when i want to prove a statement about “all ...”

Proof

“Any element occuring in the roots of a tree must occur in some leaf.”

Proof by infuction

- 1 | **Base case**
- 2 | | tree is a single leaf
- 3 | | the root of the tree is the whole tree, so it must occur in (the only) leaf
- 4 | **Inductive step**
- 5 | | Assume that for any statement aRb
- 6 | | then a must come from $*$ (the left subtree) and b must come from $+$ (the right subtree)